

Lab 3 Activity - Divide and Conquer

Part 1: RSA

Note: Look online for a list of primes, choose primes p, q, e that are > 100

1. Write primes p and q (> 100):

2. Compute $n = p \cdot q$ and $\phi = (p-1)(q-1)$

3. Choose a prime $e < \phi$ that is coprime to ϕ

Make sure that $\phi \% e \neq 0$

Write Public Key (n, e):

4. Use the **ModInv** java program to compute the private key

Compute d such that $(e \cdot d \bmod \phi == 1)$

Private Key (n, d):

5. Give your public key to another student

6. Choose a message m (an int $< n$)

7. Encrypt your message with another student's public key

Use **ModExp** program to compute $c = m^e \bmod n$

8. Give the encrypted message back to the original student

9. Decrypt the message you received with your private key

Use **ModExp** program to compute $c^d \bmod n$

10. Write the answer here. Verify with the original student